Supplementary Material

This Supplementary Material provides detailed information on the data processing, narrative generation, and model evaluation steps of the ClaMPAPP pipeline. The document is structured as follows:

- **Feature Extraction (Notes S1–S2):** Details the structured schema and validation constraints applied to clinical narratives.
- **Narrative Generation (Notes S3–S7):** Explains the templating, prompt design, resulting excerpts, fidelity audits, and permutation techniques used to convert tabular data into fluent text.
- **Model Evaluation (Notes S8–S9):** Presents comprehensive baseline machine learning metrics and probability calibration results for both internal and external cohorts.

S1: Feature Extraction Schema

The following table details the structured feature schema extracted from the clinical narratives by the `extract_features` tool in the ClaMPAPP pipeline.

Table S1: Structured feature schema extracted from free-text clinical narratives by the `extract_features` tool in ClaMPAPP.

Category	Extracted variables
Demographics and anthropometrics	Age, Sex, BMI, Height, Weight
Clinical scores	Alvarado_Score, Pediatric_Appendicitis_Score
Symptoms and examination findings	Migratory_Pain, Lower_Right_Abd_Pain, Contralateral_Rebound_Tenderness, Ipsilateral_Rebound_Tenderness, Coughing_Pain, Nausea, Loss_of_Appetite, Dysuria, Stool, Peritonitis, Psoas_Sign, Body_Temperature
Laboratory variables	WBC_Count, Neutrophil_Percentage, Neutrophilia, RBC_Count, Hemoglobin, RDW, Thrombocyte_Count, CRP, Ketones_in_Urine, RBC_in_Urine, WBC_in_Urine
Ultrasound and encounter-related variables	US_Performed, US_Number, Appendix_on_US, Appendix_Diameter, Free_Fluids, Length_of_Stay

S2: Feature Validation Constraints

As part of the ClaMPAPP pipeline, a deterministic *Feature Validator* acts as a safety gate between the LLM extractor and the XGBoost classifier. Its primary role is to intercept hallucinated or physiologically impossible values generated by the LLM. It enforces predefined clinical bounds, such as:

- $0 \leq \text{Age} \leq 18$

- $35.0 \leq$ Temperature ≤ 42.0°C
- $3.0 \leq$ WBC ≤ 50.0 (in units of 10^9/L)

Rather than clipping invalid values to the nearest boundary, any violated constraint is strictly mapped to a missing value (`NaN`).

Furthermore, to prevent the LLM from probabilistically inferring clinical scores from partial data, the Alvarado Score and the Pediatric Appendicitis Score (PAS) are calculated deterministically. They are computed only when all of their constituent variables are successfully extracted from the text; in cases of incomplete information, these scores are not computed.

S3: Patient Case Narrative Template

For the first two stages of the narrative-generation pipeline, we defined a fixed template to convert each patient's tabular record into a structured English case narrative written in the first person. For each patient in the Regensburg and Düsseldorf cohorts, the corresponding tabular values were inserted into this template. The complete template and generation code will be made publicly available upon publication.

The following is an excerpt of this fixed template used to convert each patient's tabular record into a structured English case narrative:

Patient Case Narrative Template (Excerpt)

```
I am {Age} years old with a BMI of {BMI}, identifying as {Sex}. I stand {Height}
cm tall and weigh {Weight} kg.
My Alvarado score was {Alvarado_Score} and my pediatric appendicitis score was
{Pediatric_Appendicitis_Score}.
On ultrasound, the appendix was noted as {Appendix_on_US}, measuring
{Appendix_Diameter} mm in diameter.
I reported migratory pain: {Migratory_Pain}, as well as right lower quadrant
pain: {Lower_Right_Abd_Pain}, contralateral rebound tenderness:
{Contralateral_Rebound_Tenderness}, and pain upon coughing: {Coughing_Pain}.
I experienced nausea: {Nausea} and loss of appetite: {Loss_of_Appetite}. My
temperature was {Body_Temperature} °C.
…
```

S4: Narrative Rewriting Prompt

Because templated narratives are stylistically rigid, the third stage of our pipeline utilized the Llama-3.1-8B model to rewrite the text into fluent clinical notes. The model was heavily constrained to strictly preserve all clinical facts without adding external interpretation.

The Llama-3.1-8B model was instructed to rewrite each structured template into fluent first-person English using the following strict prompt:

Narrative Rewriting Prompt

```
You are a professional medical-text editor. Your only job is to rewrite the
user's English text about a case related to appendicitis. Follow these rules
exactly:

1. ONLY rewrite the provided text to make it sound natural and human-written.

2. PRESERVE every clinical feature, sign, symptom, and factual detail. Do NOT
   remove, change, or add any clinical facts.

3. Do NOT provide explanations, diagnoses, or commentary.

4. Return EXACTLY one rewritten text block and NOTHING else.

5. If the original text contains contradictions, keep the original medical
   content but rewrite the phrasing to be natural.
```

S5: LLM-Rewritten Case Narrative Excerpt

The following excerpt illustrates the output generated by the LLM after processing the templated narrative:

Excerpt of an LLM-Rewritten Case Narrative

I'm 12 years and 8 months old, with a body mass index of 16.9. I'm female and 148 centimeters tall, weighing 37 kilograms. …I experienced pain in my lower right abdomen, which I described as a migratory pain, and had tenderness on the opposite side of my body when I pressed. …Laboratory tests showed 7.7 white blood cells per microliter, and my CRP level was zero. …An ultrasound of my abdomen revealed that my appendix measured 7.1 millimeters in diameter and showed signs of being inflamed. …

S6: Fidelity Audit of Narrative Rewriting

Because LLM outputs may be sensitive to instruction phrasing, information order, and information quantity, and may not reliably follow instructions [1], we performed a manual fidelity audit to compare the rewritten narratives against the original tabular ground truth. A random sample of 31 clinical reports was drawn from the Regensburg evaluation split, corresponding to 15.35% of the $N = 202$ internal evaluation cohort. For each report, we verified the exact preservation of approximately 20 core clinical features, including demographics, vital signs, laboratory values, physical examination findings, imaging findings, and clinical scores.

At the report level, the 31 sampled cases demonstrated high fidelity: 14 reports showed perfect fidelity, 13 contained exactly one feature-level deviation, and 4 contained exactly two feature-level deviations. No report contained more than two deviations. Across the audited sample, corresponding to approximately 620 feature-level checks, we identified 21 isolated feature-level errors, equivalent to 3.4%. These deviations comprised feature omissions ($n = 10$), contradictions ($n = 7$), qualitative abstractions ($n = 3$), and one null imputation ($n = 1$), in which a missing value, `NaN`, was rendered as zero in the generated text.

To assess generalizability across datasets, we repeated the audit on a random sample of 10 reports from the Düsseldorf cohort. This audit covered 17 features, corresponding to approximately 170 feature-level checks. The results were similar: 7 reports showed perfect fidelity, 1 report contained exactly one deviation, and 2 reports contained exactly two deviations. At the feature level, we observed five deviations: three omissions, one contradiction, and one qualitative abstraction, in which a tabular "no" was paraphrased as "normal".

Overall, the fidelity audits indicated that the rewriting stage introduced natural linguistic variation while maintaining a low rate of content drift. With feature-level error rates of 3.4% for Regensburg and 2.9% for Düsseldorf, the pipeline preserved more than 96% of the audited information across both cohorts, supporting the use of the rewritten narratives as a high-fidelity proxy for downstream feature-extraction experiments.

S7: Sentence Permutation and Structural Inversion

To systematically assess the robustness of our feature-extraction pipeline to variations in information ordering, we generated permuted versions of the clinical narratives using a bipartite structural inversion technique. Each templated narrative was divided into two sequential halves. The sequential order of these two sections was then reversed before being processed by the LLM rewriting stage.

This procedure deliberately altered the absolute and relative positions of clinical facts within the text while preserving the overall coherence of the narrative segments. It provided a structured way to test ordering effects without introducing the severe semantic degradation and ungrammaticality associated with fully random sentence shuffling. This permutation methodology was consistently applied to records in both the Regensburg and Düsseldorf evaluation cohorts.

S8: Baseline ML Model Evaluation

To establish a robust classification engine for our hybrid system, we benchmarked several supervised machine learning algorithms. Detailed performance metrics were recorded for both the internal Regensburg validation set (Table S2) and the external Düsseldorf validation set (Table S3).

Tables S2 and S3 show a clear decrease in performance when models were evaluated on the external cohort, consistent with reduced transportability under cohort shift. For instance, although Random Forest achieved the highest internal specificity, both its accuracy and specificity decreased substantially on the external set. The Decision Tree model showed a similar pattern of performance drop. These results suggest weaker external generalization for the single-tree and bagging-based models, justifying the selection of the Gradient Boosting framework (XGBoost) for the primary ClaMPAPP architecture.

Table S2: Performance metrics on the internal validation set (TEST 1).

Model Name	Acc	Sen	Spec	Prec	F1	TP	TN	FP	FN
LightGBM	0.815	0.988	0.687	0.700	0.820	84	79	36	1
XGBoost	**0.895**	**0.988**	**0.826**	**0.808**	**0.889**	**84**	**95**	**20**	**1**
Decision Tree	0.870	0.906	0.843	0.811	0.856	77	97	18	8
Random Forest	0.890	0.824	0.939	0.909	0.864	70	108	7	15
Logistic Regression	0.785	0.871	0.722	0.698	0.775	74	83	32	11

S9: Probability Calibration of ClaMPAPP

Because ClaMPAPP outputs continuous appendicitis risk scores, we evaluated not only discrimination but also probability calibration on both cohorts. On the internal Regensburg cohort, ClaMPAPP showed excellent discrimination (AUC-ROC = 0.982), but the raw predicted probabilities were not perfectly calibrated. The raw Brier score was 0.118, log loss was 0.375, and the expected calibration error (ECE; 10 bins) was 0.167. The calibration intercept and slope were -2.936 and 1.113, respectively, and the Hosmer–Lemeshow test indicated lack of perfect

Table S3: Performance metrics on the external validation set (TEST 2).

Model Name	Acc	Sen	Spec	Prec	F1	TP	TN	FP	FN
LightGBM	0.807	0.935	0.394	0.833	0.881	215	28	43	15
XGBoost	**0.811**	**0.917**	**0.465**	**0.847**	**0.881**	**211**	**33**	**38**	**19**
Decision Tree	0.738	0.817	0.479	0.836	0.826	188	34	37	42
Random Forest	0.767	0.874	0.423	0.831	0.852	201	30	41	29
Logistic Regression	0.781	0.939	0.268	0.806	0.867	216	19	52	14

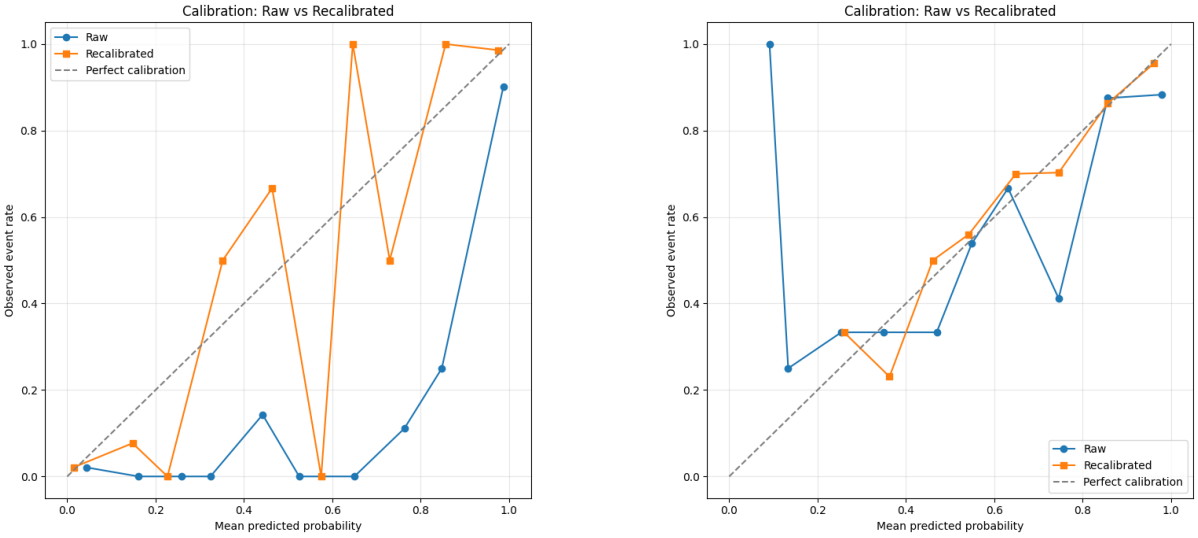
fit ($p < 0.001$). Thus, although the model ranked cases very effectively, the raw probability scale showed imperfect agreement with observed event frequencies.

On the external Düsseldorf cohort, raw calibration was also imperfect. The AUC-ROC was 0.804, the Brier score was 0.150, log loss was 0.508, and ECE was 0.096. The calibration intercept was close to zero (-0.060), whereas the calibration slope was substantially below 1 (0.513), indicating that the spread of predicted probabilities was misaligned with observed outcome frequencies. The Hosmer–Lemeshow test again suggested lack of perfect fit ($p < 0.001$). Reliability diagrams for both cohorts are shown in Figure 1.

In an exploratory post hoc analysis, logistic recalibration substantially improved *apparent* calibration metrics on both cohorts (Table S4). These findings suggest that ClaMPAPP's miscalibration may be due, at least in part, to a mismatch in the probability scale rather than poor ranking ability, and that this mismatch can be improved by simple logistic recalibration. Because recalibration and evaluation were performed on the same cohorts, these results reflect apparent rather than externally validated calibration improvement.

Table S4: Calibration metrics for ClaMPAPP on the internal Regensburg and external Düsseldorf cohorts. Recalibrated values are from exploratory post hoc logistic recalibration performed and evaluated on the same cohort; therefore, these results reflect *apparent* calibration improvement.

Cohort	Metric	Raw	Recalibrated
Regensburg	ROC AUC	0.982	0.982
	Brier score	0.118	0.046
	Log loss	0.375	0.164
	ECE (10 bins)	0.167	0.039
	Calibration intercept	-2.936	≈ 0
	Calibration slope	1.113	≈ 1
Düsseldorf	ROC AUC	0.804	0.804
	Brier score	0.150	0.140
	Log loss	0.508	0.431
	ECE (10 bins)	0.096	0.026
	Calibration intercept	-0.060	≈ 0
	Calibration slope	0.513	≈ 1



(a) Regensburg cohort

(b) Düsseldorf cohort

Figure 1: **Calibration of ClaMPAPP on the internal and external cohorts.** Reliability diagrams compare observed event rates with mean predicted probabilities across bins. **(A) Regensburg cohort:** raw probabilities show marked deviation from the ideal diagonal despite excellent discrimination; post hoc logistic recalibration improves apparent agreement between predicted and observed risk. **(B) Düsseldorf cohort:** raw probabilities also show imperfect calibration, particularly in the spread of predicted risks; recalibration again improves apparent calibration. In both cohorts, recalibration leaves ranking performance unchanged and primarily adjusts the interpretation of scores as absolute risk probabilities.

References

[1] Hager, P., Jungmann, F., Holland, R., Bhagat, K., Hubrecht, I., Knauer, M., Vielhauer, J., Makowski, M., Braren, R., Kaissis, G., & Rueckert, D. Evaluation and Mitigation of the Limitations of Large Language Models in Clinical Decision-Making. *Nature Medicine*, 30(9), 2613–2622 (2024). https://doi.org/10.1038/s41591-024-03097-1